

Assignment-7

1. In cryptography, a Caesar cipher is a very simple encryption techniques in which each letter in the plain text is replaced by a letter some fixed number of positions down the alphabet. For example, with a shift of 3, A would be replaced by D, B would become E, and so on. The method is named after Julius Caesar, who used it to communicate with his generals. ROT-13 ("rotate by 13 places") is a widely used example of a Caesar cipher where the shift is 13. In Python, the key for ROT-13 may be represented by means of the following dictionary:

- `key = {'a':'n', 'b':'o', 'c':'p', 'd':'q', 'e':'r', 'f':'s', 'g':'t', 'h':'u', 'i':'v', 'j':'w', 'k':'x', 'l':'y', 'm':'z', 'n':'a', 'o':'b', 'p':'c', 'q':'d', 'r':'e', 's':'f', 't':'g', 'u':'h', 'v':'i', 'w':'j', 'x':'k', 'y':'l', 'z':'m', 'A':'N', 'B':'O', 'C':'P', 'D':'Q', 'E':'R', 'F':'S', 'G':'T', 'H':'U', 'I':'V', 'J':'W', 'K':'X', 'L':'Y', 'M':'Z', 'N':'A', 'O':'B', 'P':'C', 'Q':'D', 'R':'E', 'S':'F', 'T':'G', 'U':'H', 'V':'I', 'W':'J', 'X':'K', 'Y':'L', 'Z':'M'}`

- Your task in this exercise is to implement an encoder/decoder of ROT-13. Once you're done, You will be able to read the following secret message:

- **Pnrfne pvcure? V zhpu cersre Pnrfne fnynq!**

- **Note:** that since English has 26 characters, your ROT-13 program will be able to both encode And decode texts written in English

2. Write a version of a palindrome recognizer that also accepts phrase palindromes such as

- "Go hanga salami I'm a lasagna hog.",
- "Was it a rat I saw?",
- "Step on no pets",
- "Sit on a potato pan, Otis",
- "Lisa Bonet ate no basil",
- "Satan, oscillate my metallic sonatas",
- "I roamed under it as a tired nude Maori",
- "Rise to vote sir", or the exclamation "Dammit, I'm mad!".

- **Note:** that punctuation, capitalization, and spacing are usually ignored.

3. Write a program in Python to read a string from the keyboard. Replace each '**This**' word with '**That**' word in this String.

- Example:
 - **Input:** This is me and This is my python program.
 - **Output:** That is me and That is my python program

4. Write a program that reads a string from keyboard and display:

- a) The number of uppercase letters in the string
- b) The number of lowercase letters in the string
- c) The number of digits in the string
- d) The number of whitespace characters in the string

5. Write a function in python to count the number of lines from a text file "story.txt" which is not starting with an alphabet "C".

- **Example:** If the file "story.txt" contains the following lines:

- Connecting DB's with Python.
- Working with DB's using Python.
- Accessing and Manipulating DB's.
- Creation of Python virtual Environment.
- Working with beautiful soup.
- Working with matplotlib, seaborn.
- The function should display the **output** as **4**

6. Write a function in Python to count and display the total number of words in a text file.

- **Note:** use above "**story.txt**" file

